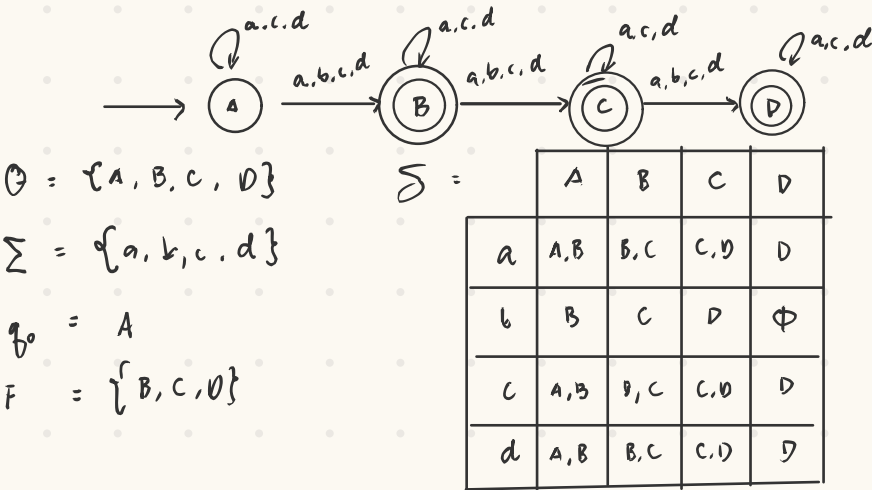
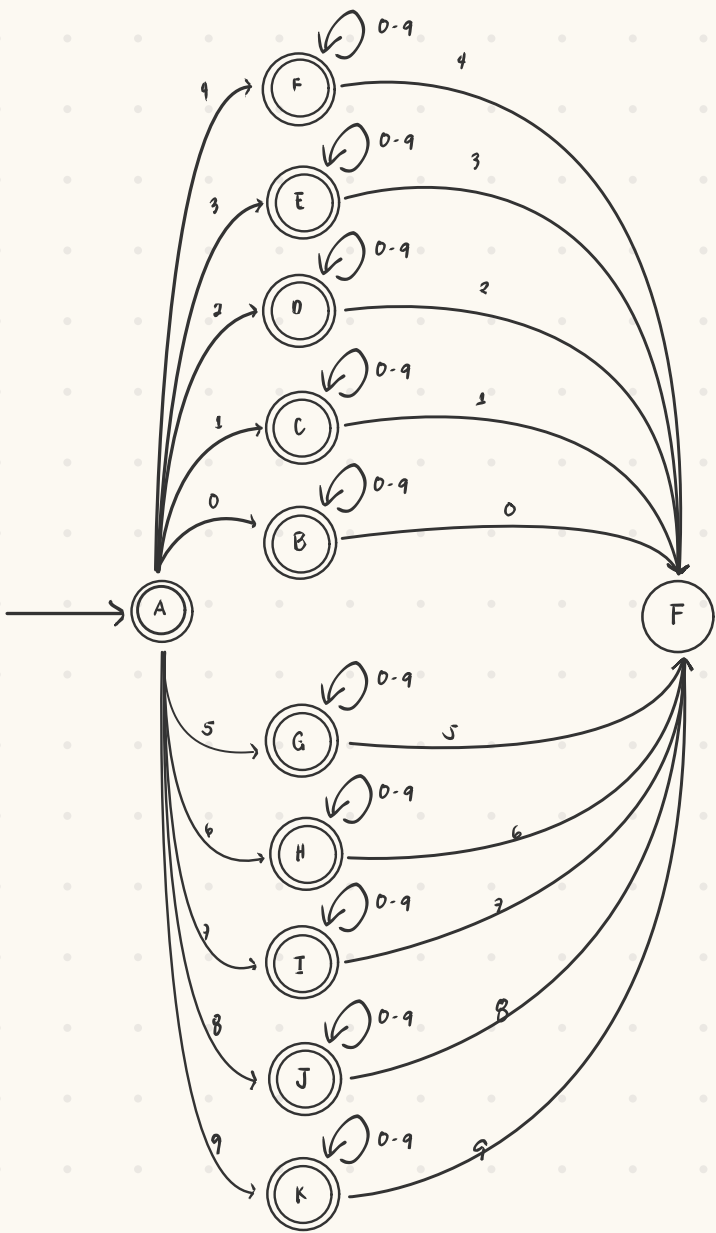


Part 1:

#01 Strings over {a,b,c,d} with at most three 'b's



#02 Strings over {0,1,2,...,9} such that the final digit has not appeared before



$Q = \{A, B, C, D, E, F, G, H, I, J, K, L\}$

$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$q_0 = L$

$F = \{A, B, C, D, E, F, G, H, I, J, K\}$

$\delta = Q \times \Sigma = 2^a$

Design an q -NFA that recognizes the language

3. The set of strings consisting of zero or more 'a's followed by zero or more 'b's, followed by zero or more 'c's

